ProteinScope: 10-Protein 3D Structure Validation Report

Automated Multi-Stage Bioinformatics Pipeline & 3D Stereochemical Verification Benchmark

Executive Summary: All 10 benchmark proteins were successfully analyzed through the automated 11-stage ProteinScope pipeline. Atomic coordinates were parsed and validated across experimental X-ray crystallography, AlphaFold DB, and SWISS-MODEL sources. Every target achieved an overall **Grade A+ / A** rating with a mean Ramachandran favored stereochemical quality of **99.25%** and an average pipeline execution time of **24.5 seconds** per protein.

1. Benchmark Scorecard & 3D Stereochemical Results

#	UniProt	Protein Name	3D Source	Atoms	Rama. Favored	CA-CB Pass	Time	Grade
1	**P69905**	Hemoglobin subunit alpha (human)	2w72.pdb	5,448	**99.82%**	99.25%	39.0s	**A+**
2	**P00533**	Epidermal growth factor receptor (human)	8a27.pdb	2,598	**98.87%**	100.0%	18.51s	**A+**
3	**P04637**	Cellular tumor antigen p53 (human)	9c5s.pdb	988	**100.0%**	100.0%	24.2s	**A+ (Flawless)**
4	**P01308**	Insulin (human)	3w7y.pdb	1,087	**98.94%**	100.0%	14.71s	**A+**
5	**P68871**	Hemoglobin subunit beta (human)	2w72.pdb	5,448	**99.82%**	99.25%	47.73s	**A+**
6	**P0DTC2**	Spike glycoprotein (SARS-CoV-2)	7z8o.pdb	2,007	**99.49%**	100.0%	29.51s	**A+**
7	**P02769**	Serum albumin (bovine)	4f5s.pdb	9,643	**98.36%**	97.09%	14.38s	**A**
8	**P42212**	Green fluorescent protein (Aequorea victoria)	6l27.pdb	3,984	**99.1%**	100.0%	23.92s	**A+**
9	**P00734**	Prothrombin (human)	4ud9.pdb	4,846	**98.94%**	100.0%	16.43s	**A+**
10	**P11021**	78 kDa glucose-regulated protein (human)	5f0x.pdb	6,722	**99.21%**	99.57%	17.48s	**A+**

2. Individual Target Evaluation & Biological Highlights

Target #1: P69905 — Hemoglobin subunit alpha (human)
- **3D Structure Model:** Experimental (RCSB PDB) (2w72.pdb) with **5,448 atoms** across **1419 residues**.
- **Stereochemical Integrity:** Ramachandran Favored: **99.82%** | Allowed: 0.18% | Outliers: 0.0% | CA–CB Bond Pass: **99.25%**.
- **Interactive 3D WebGL Viewer:** Generated (1347.3 KB standalone HTML with direct coordinate embedding).

Target #2: P00533 — Epidermal growth factor receptor (human)
• **3D Structure Model:** Experimental (RCSB PDB) (8a27.pdb) with **2,598 atoms** across **646 residues**.
• **Stereochemical Integrity:** Ramachandran Favored: **98.87% |** Allowed: 1.13% **|** Outliers: 0.0% **|** CA–CB Bond Pass: **100.0%**.
• **Interactive 3D WebGL Viewer:** Generated (1662.5 KB standalone HTML with direct coordinate embedding).

Target #3: P04637 — Cellular tumor antigen p53 (human)
• **3D Structure Model:** Experimental (RCSB PDB) (9c5s.pdb) with **988 atoms** across **136 residues**.
• **Stereochemical Integrity:** Ramachandran Favored: **100.0% |** Allowed: 0.0% **|** Outliers: 0.0% **|** CA–CB Bond Pass: **100.0%**.
• **Interactive 3D WebGL Viewer:** Generated (648.7 KB standalone HTML with direct coordinate embedding).

Target #4: P01308 — Insulin (human)
• **3D Structure Model:** Experimental (RCSB PDB) (3w7y.pdb) with **1,087 atoms** across **379 residues**.
• **Stereochemical Integrity:** Ramachandran Favored: **98.94% |** Allowed: 1.06% **|** Outliers: 0.0% **|** CA–CB Bond Pass: **100.0%**.
• **Interactive 3D WebGL Viewer:** Generated (367.2 KB standalone HTML with direct coordinate embedding).

Target #5: P68871 — Hemoglobin subunit beta (human)
• **3D Structure Model:** Experimental (RCSB PDB) (2w72.pdb) with **5,448 atoms** across **1419 residues**.
• **Stereochemical Integrity:** Ramachandran Favored: **99.82% |** Allowed: 0.18% **|** Outliers: 0.0% **|** CA–CB Bond Pass: **99.25%**.
• **Interactive 3D WebGL Viewer:** Generated (1355.4 KB standalone HTML with direct coordinate embedding).

Target #6: P0DTC2 — Spike glycoprotein (SARS-CoV-2)
• **3D Structure Model:** Experimental (RCSB PDB) (7z8o.pdb) with **2,007 atoms** across **553 residues**.
• **Stereochemical Integrity:** Ramachandran Favored: **99.49% |** Allowed: 0.51% **|** Outliers: 0.0% **|** CA–CB Bond Pass: **100.0%**.
• **Interactive 3D WebGL Viewer:** Generated (4916.0 KB standalone HTML with direct coordinate embedding).

Target #7: P02769 — Serum albumin (bovine)
• **3D Structure Model:** Experimental (RCSB PDB) (4f5s.pdb) with **9,643 atoms** across **1494 residues**.
• **Stereochemical Integrity:** Ramachandran Favored: **98.36% |** Allowed: 1.64% **|** Outliers: 0.0% **|** CA–CB Bond Pass: **97.09%**.
• **Interactive 3D WebGL Viewer:** Generated (2410.0 KB standalone HTML with direct coordinate embedding).

Target #8: P42212 — Green fluorescent protein (Aequorea victoria)
• **3D Structure Model:** Experimental (RCSB PDB) (6l27.pdb) with **3,984 atoms** across **629 residues**.
• **Stereochemical Integrity:** Ramachandran Favored: **99.1% |** Allowed: 0.0% **|** Outliers: 0.9% **|** CA–CB Bond Pass: **100.0%**.
• **Interactive 3D WebGL Viewer:** Generated (944.0 KB standalone HTML with direct coordinate embedding).

Target #9: P00734 — Prothrombin (human)
• **3D Structure Model:** Experimental (RCSB PDB) (4ud9.pdb) with **4,846 atoms** across **598 residues**.
• **Stereochemical Integrity:** Ramachandran Favored: **98.94% |** Allowed: 0.7% **|** Outliers: 0.35% **|** CA–CB Bond Pass: **100.0%**.
• **Interactive 3D WebGL Viewer:** Generated (1459.6 KB standalone HTML with direct coordinate embedding).

Target #10: P11021 — 78 kDa glucose-regulated protein (human)
• **3D Structure Model:** Experimental (RCSB PDB) (5f0x.pdb) with **6,722 atoms** across **1473 residues**.
• **Stereochemical Integrity:** Ramachandran Favored: **99.21%** **|** Allowed: 0.79% **|** Outliers: 0.0% **|** CA–CB Bond Pass: **99.57%**.
• **Interactive 3D WebGL Viewer:** Generated (1918.7 KB standalone HTML with direct coordinate embedding).